PawWatch

Edge-AI Cat Health Monitor

GitHub Repo

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Problem

Cat hides illness

Cats instinctively hide pain and illness; obvious symptoms such as vomiting may appear only when it is more severe [1], [2]. Earlier signs show up in daily habits: vets list changes in appetite, thirst and litter box use as warning signs [2], and each common disease changes them in its own way:

Disease	How common	Eating	Drinking	Litter
Kidney disease	Up to 30–40% of older cats may have it; only 3.6% are diagnosed [3], [4]		↑	↑
Hyperthyroidism	8.7% of cats over 10 [5], [6]	↑	↑	↑
Diabetes	About 1 in 200 cats [7]	↑	↑	↑
Urinary tract disease	A leading reason cats visit vets [8]			↑

A 2026 study of 136 cats found those with kidney disease used the litter box about twice as often, even in early stages [9]. A blocked urinary tract may kill within 24–48 hours; its first sign is many litter box trips [8].

Solution

Today’s smart devices each watch one spot

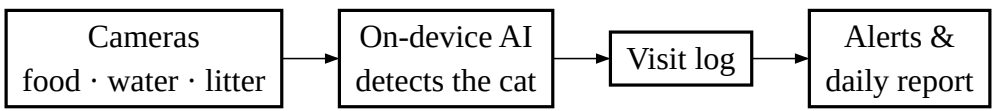
Each smart device on the market watches one spot, so watching food, water and litter takes several devices and apps.

	Food	Water	Litter
Smart feeders	✓	✗	✗
Smart water fountains	✗	✓	✗
Smart litter boxes	✗	✗	✓
Pet cameras (video only, no visit log)	✗	✗	✗
PawWatch	✓	✓	✓

However, illness shows up as a **combination**. For example, kidney disease raises both drinking and litter box use [3], so a water-only or litter-only device sees half the pattern. Once set up, PawWatch watches all three spots with one box, and the video never leaves home.

Solution: PawWatch

Owners are out all day, and no single device sees the whole pattern. **PawWatch** is a vision-based monitoring system: low-cost cameras watch the food bowl, water bowl and litter box, and an on-device AI accelerator recognizes the cat and logs every visit. Video never leaves home.



What it records

- **Visit log:** how often and how long the cat eats, drinks and uses the litter box
- **No-go zone alarm:** plays a sound when the cat jumps onto the counter

What it tells you

- **Anomaly alert:** ≥ 2 or $\leq \frac{1}{2}$ times of the cat's 7-day average
- **Daily report:** a plain-language summary
- **Dashboard:** today's counts, timeline and 7-day trends

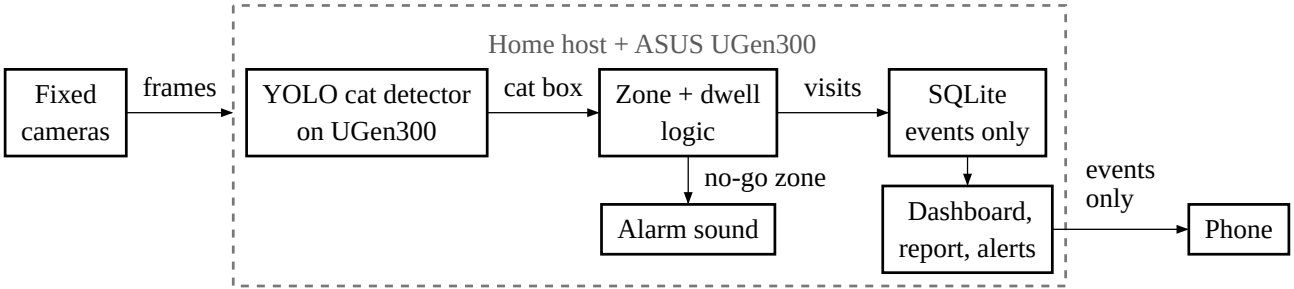
Why run AI at home, not in the cloud

	Cloud pet camera	PawWatch
Who sees your home video?	It's uploaded to the company's servers. At one home-camera maker, staff watched customers' private videos [10]	No one. Video never leaves home; only events like "14:32, drank for 25 s" are stored
Monthly fee	Smart features usually need a paid subscription	None. No servers to pay for
Internet goes down	Cloud features stop until it's back	Keeps logging; alerts are sent once it's back
Cat jumps on the counter?	Frames go to the server and back first	The alarm sounds right away, from the box at home

Running 24/7 is cheap: the AI accelerator draws 2.5 W, less than an LED bulb [11].

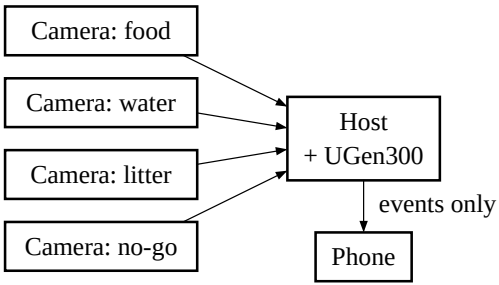
PawWatch System

Architecture: video in, events out



- **Detect:** the host decodes each camera's video and the UGen300 runs a stock YOLO model that finds the cat. Frames live only in memory and are dropped after inference; nothing saves or uploads them.
- **Turn boxes into visits:** the cat's box is matched to the zones drawn for that camera (food, water, litter, no-go), and a visit is logged once the cat stays long enough. A no-go zone rings the alarm right away.
- **Store and report:** each visit is one row in a local SQLite database. The dashboard, anomaly alert and daily report read only these rows, so the only thing that can leave the home is an event like "07:30, ate for 31 s".

Hardware: one host, many fixed cameras



We mount **fixed cameras**, one at each spot we watch: the food bowl, the water bowl, the litter box and any no-go spot such as the kitchen counter. Each camera streams its video over the home network to **one small host computer** (a mini PC or Raspberry Pi).

The **ASUS UGen300** is plugged into the host over USB or M.2. The host decodes the video and hands frames to the UGen300, whose Hailo-10H chip (40 TOPS at 2.5 W [11]) runs the cat detector. The host then turns detections into visits, stores them and serves the dashboard. Only these events reach the owner's phone.

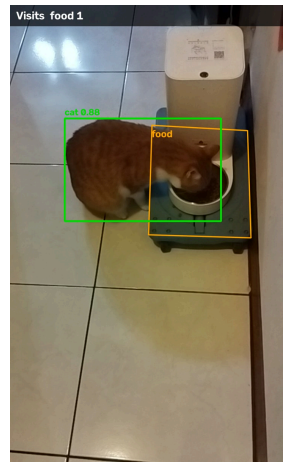
Why fixed cameras: zones are drawn in pixels, so a camera that never moves needs them drawn only once, at setup. One camera per spot also keeps the cat large in the frame, which makes detection reliable.

Detection: from pixels to “the cat is at the bowl”

A stock YOLO model, no training. We use YOLOv8n pretrained on the public COCO dataset, which already has a “cat” class. We keep only that class (score ≥ 0.4). No cat photos were collected and no model was trained or fine-tuned, so any home can use it out of the box.

Zones are drawn once. At setup, the owner clicks a polygon around each spot on a still frame (`draw_zones.py`); it is saved as one small JSON file per camera.

Which zone is the cat in? We take the highest-scoring cat box and measure how much of it overlaps each zone, as a share of the smaller of box and zone. The zone with the most overlap wins, if it reaches 20%. Measuring against the smaller shape lets a zone hug a small bowl, and a box that only grazes a zone’s edge doesn’t count.



Detection window: cat box, food zone and visit count

Event logic: from detections to visits

The event logic is pure code: it takes $(\text{video time}, \text{zone})$ pairs and returns visits, so it is tested without a model or video. Timestamps are **video time** (recording start + frame offset), so recorded footage replays exactly.



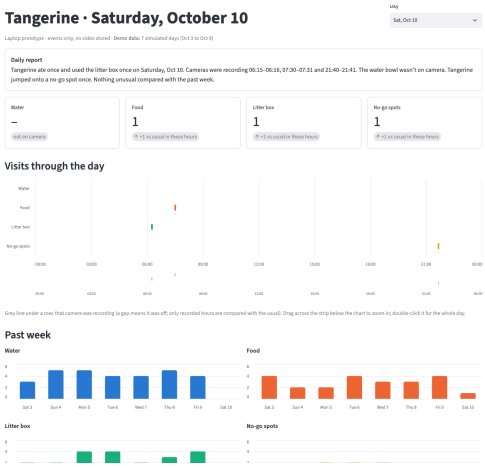
Minimum dwell	A visit counts after 5 s in one zone, so walking past a bowl isn't a meal. No-go zones need only 1 s: a quick jump still counts.
Gap tolerance	The cat may vanish for up to 5 s (turning, hidden by the litter box rim) without splitting one visit into two.
Live writes	A visit is stored as soon as it qualifies and its end moves forward while the cat stays, so the dashboard follows a live camera.
Alarm	Every frame in a no-go zone calls the alarm at once; a 30 s cooldown gives one sound per jump, not per frame.

Storage and outputs: what the owner sees

One table is the whole record. Each visit is one row in local SQLite: `camera, zone, start, end, duration, simulated`. A second table logs when each camera was recording, so hours with the camera off are never read as “didn’t drink”.

Anomaly alert. For each of food, water and litter, today’s count is compared with the cat’s own average over the previous 7 days, over the same hours, so a normal morning isn’t flagged for a missing dinner. An alert needs both ≥ 2 or ≤ 0.5 times the average and a difference of at least 3 visits, so 1 vs. 0 is not an alert.

Daily report and alarm. Template sentences turn the counts into plain language. Wording suggests watching the cat or seeing a vet, never a diagnosis. No-go jumps play a sound on the host and can send a text message (never a frame).



Dashboard: today’s counts, timeline, 7-day trend

Impact and Business

Benefits for the cat: problems caught early, a safer home

Catch problems early

- **Early signs of many diseases**: appetite, thirst and litter use change before obvious symptoms (see Problem, p. 3), and PawWatch tracks all three every day
  Example: kidney disease doubles litter use even at early stages [9], yet most cases go undiagnosed [3], [4]. Found early, it is treatable: a kidney diet cut uremic crises from 26% to 0% in a 45-cat trial [12]
- **Emergencies in hours, not days**: an urgent change shows up in that evening's report, not after days of guessing
  Example: a blocked urethra can kill within 24–48 hours [8]; a cat that stops eating for a few days can develop fatty liver disease, which can be fatal [13]

A safer, calmer home

- **No-go zones**: an alarm scares the cat off, and the owner gets a LINE or Discord text right away
  Example: a cat that eats a lily needs treatment within 18 hours, or its kidney failure is generally irreversible [14]
- **Less stress**: the cat is observed at home, unhandled, so the record shows its normal behavior
  Example: a simulated clinic visit changed healthy cats' blood pressure by up to 80 mmHg [15]

Benefits for the owner: rest assured, wherever you are

- **Peace of mind:** at work, on a business trip or in hospital, the cat is watched all day, not once a day like a pet sitter visit. Alerts reach the phone over LINE or Discord
- **Facts instead of guesses:** vet guidelines value accurate daily water records over the owner's impression [16]; 55.5% of owners did not recognize more thirst as a sign of dehydration (manufacturer survey) [17]. PawWatch counts every visit
- **One box, no subscription:** single-spot smart devices cost NT$1,688–38,800 each [18], [19], [20]; cloud pet cameras charge monthly, e.g. Furbo from US$9.99 [21] (see Business model, p. 19)
- **Privacy:** video never leaves home (see Why run AI at home, p. 7)

Earlier is cheaper

Caught early
A senior cat check-up: NT$3,000–8,000 [22]

Caught late
Emergency catheterization of a blocked cat: up to NT$10,000, +50% at night, plus each day in hospital; surgery (urethrostomy): up to NT$50,000 [23]. One owner paid NT$30,000 at a 24-hour hospital [24]

Benefits for vets and beyond

Vets

- **A better history:** a 7-day trend of visits instead of “I think he drinks more” [2]
- **Chronic care at home:** for diabetic cats, guidelines call less drinking “a useful clinical marker of response to insulin therapy” and ask owners to keep a diary [16]. PawWatch keeps it automatically
- **After surgery:** track eating, drinking and litter use at home for any abnormal sign

Beyond the home

- **Low power:** the AI accelerator draws 2.5 W [11], and no video is streamed to a data center
- **Shelters and street cats:** once PawWatch tells cats apart (future work), the same system can watch shelter rooms and TNR feeding stations

Market: 266k Taiwan homes with an older cat

TAM · 1.02 M homes · **NT$10.2 B**

SAM · 266 k homes · **NT$2.7 B**

SOM · 5.3 k homes · **NT$53 M**

9.85 M households [25] × 10.4% keep cats [26]

× 26% of dogs and cats are 7 or older [27]: the age when chronic disease becomes common

× 2% of SAM reached in years 1–3 (our assumption)

- **Price:** NT$10,000 per kit, a working assumption (see Business model, p. 19)
- **Sanity check:** the SOM is about one kit per animal hospital per year (about 1,800 hospitals [28])
- **Tailwind:** 1.74 M pet cats, up 33% in two years [29]; global pet tech may grow from US$15.6 B (2025) to US$52.9 B (2035) [30]

Conservative: the household share is from 2023, before the 33% rise in cats.

Business model: ride on hardware people already buy

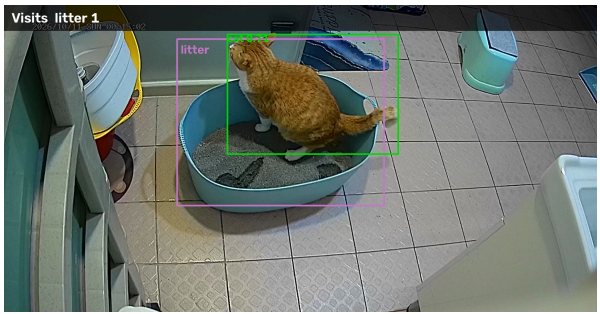
TODO: 目前是 AI 生的, 完全不知道這啥

Customer	Offer	Revenue
Device makers (pet cameras, home hubs, PCs with edge AI)	PawWatch edge software on their hardware: no cloud cost for them	Per-unit license
Vet clinics	At-home monitoring kits after surgery and for chronic cases (e.g. kidney disease)	Monthly rental per kit
Cat owners	Box + cameras, no subscription	One-time purchase

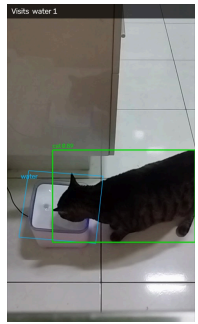
- Licensing spreads the UGen300 cost: it doesn’t land on one product
- One UGen300 serves every camera in the home, so each extra spot is only a camera

Appendix

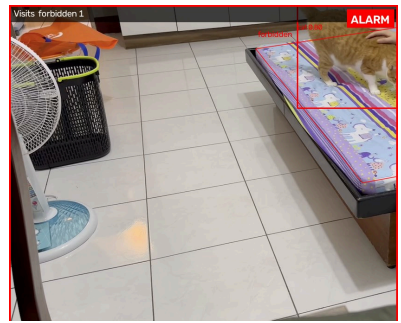
Detection screenshots



Tangerine in the litter zone



Ami at the water bowl



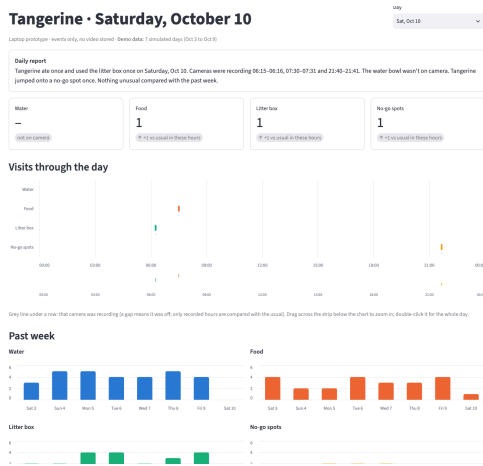
No-go zone: box turns red, alarm rings

Each window shows the cat box with its score, the camera’s zones and the visits counted so far. The windows are only displayed; frames are never written to disk.

Dashboard with an anomaly alert (simulated data)

This dashboard runs on **simulated** data only, to show what an alert looks like: a day where the litter box count reaches at least twice the cat's 7-day average.

The alert names the change and the usual count, and suggests keeping an eye on the cat or seeing a vet. It never names a disease.



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